BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

LESSON 307 · Paper 1 examiner-style feedback

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why does cache hit reduce memory delay?

Q2. What does a model answer need for a 6-mark explanation?

Q3. What is a mark-scheme method point?

Q4. Convert binary 01011000_2 to hexadecimal and decimal.

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I Do · Read new knowledge

Key vocabulary

evaluation — Judgement supported by evidence.

method mark — Award for correct reasoning step.

misconception — Incorrect conceptual model.

transfer — Applying learning to unfamiliar problem.

New learning

Diagnose rather than praise vaguely — Separate inaccurate knowledge, missing context and absent justification; use an explicit improvement target.

Show full method — Write intermediary calculations, trace states and correctly named registers or SQL keys.

Demonstrate transfer — Apply the improved method to a different question, not a copied mark-scheme answer.

Worked example

Improve 'pipelining makes instructions twice as quick'.

1. Predict first — Name the mechanism—overlapping stages of different instructions—rather than speeding an individual stage.

2. Check reasoning — State a limitation, such as dependency stalls or branch flushes.

We Do · Practise with guidance

Improve 'a JOIN combines tables' for an inner join on StudentID.

```
Student INNER JOIN Score
ON Student.StudentID=Score.StudentID
```

Predict first

Name the matching attribute and type of join rather than a generic table combination.

Check reasoning

Explain that rows with no matching StudentID in both tables are excluded.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Rewrite 'Harvard is faster' to explain architecture and qualify performance. [6]

Question 2

An answer says −6 is 11111001_2 in 8-bit two's complement. Correct it with full working. [6]

Question 3

Explain why 9-mark evaluations need balanced, contextual judgement, using a hypothetical school cloud-storage deployment. [6]

Exit Ticket · Answer before checking

Repeating the final answer without the working is always sufficient to gain method marks.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It may avoid a slower main-memory access.

Q2. Accurate mechanism, context and supported consequences.

Q3. Credit for a correct intermediate step or justified inference.

Q4. 58_{16}, decimal 88 (64+16+8).

You Do pages · Independent questions

Q1. Separate instruction/data memory pathways [2], potential simultaneous fetch/data access [2], bottleneck reduced in some designs [1], not unconditional speed advantage [1].

Q2. +6=00000110 [2], invert=11111001 [2], +1=11111010 [2]; initial answer equals −7.

Q3. Specific access benefit [1], risk of outages [1], security/privacy [1], costs [1], mitigation [1], justified conclusion relative to context [1].

Exit Ticket · Model answer

False. Working must be shown where the mark scheme requires it.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

RHS COMPUTING / OCR H446 / mixed LESSON 307 · DRAFT

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Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check understanding